

JKBOSE 2026

CLASS 11TH

#XUV 

Bhaichara
ON
TOP

ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-13

CHEMISTRY

VISHAL SIR



JKBose Wallah

Topics To Covered

1

Types of Organic Reactions

Handwritten.

2

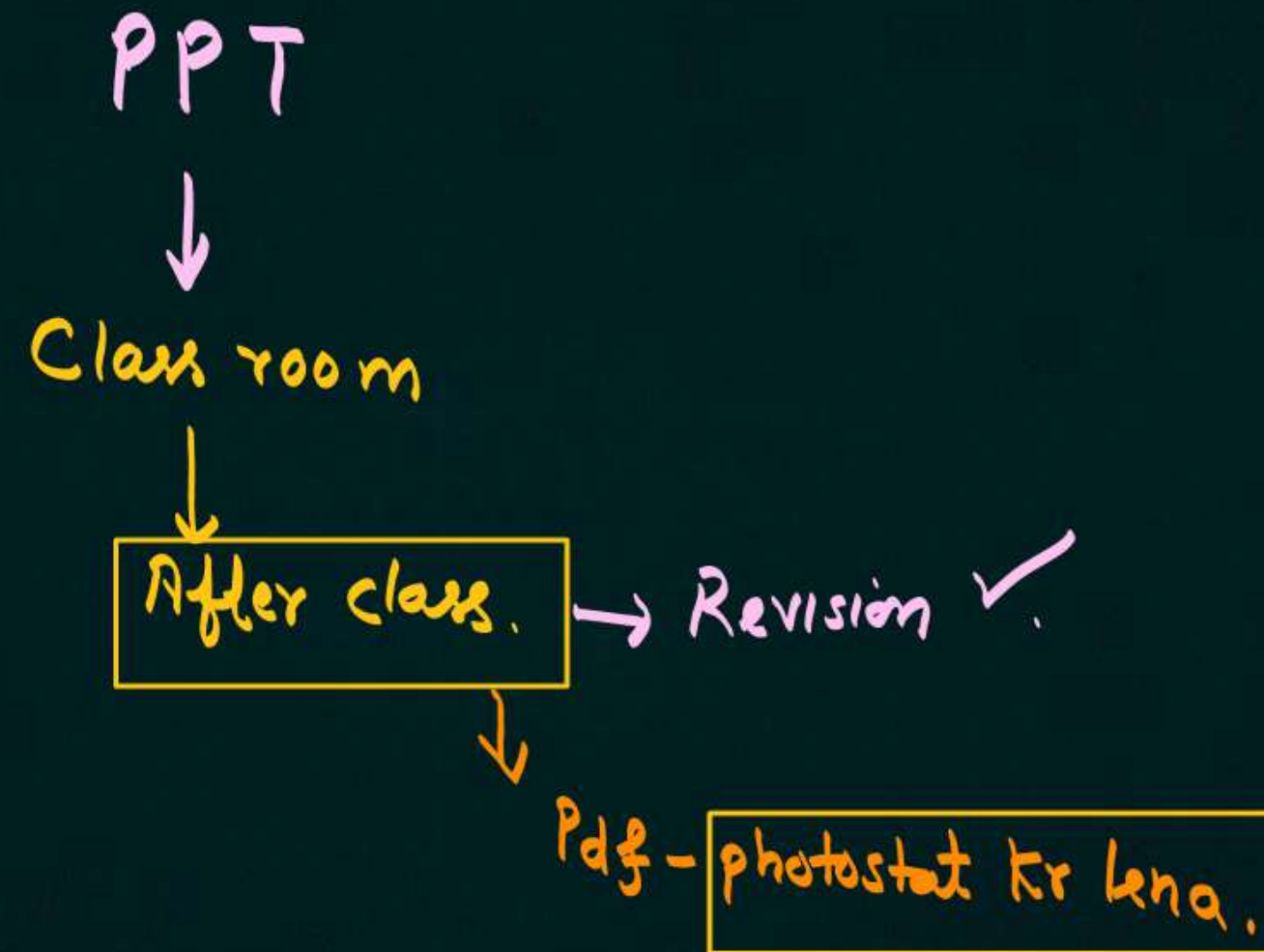
Methods of Purification of Organic Compounds

IUPAC ✓

GOC ✓

Methods ✓

lengthy.



Types of Reactions

Four Types of Organic Reactions

- (a) Substitution
- (b) Addition RxN.
- (c) Elimination RxN.
- (d) Rearrangement

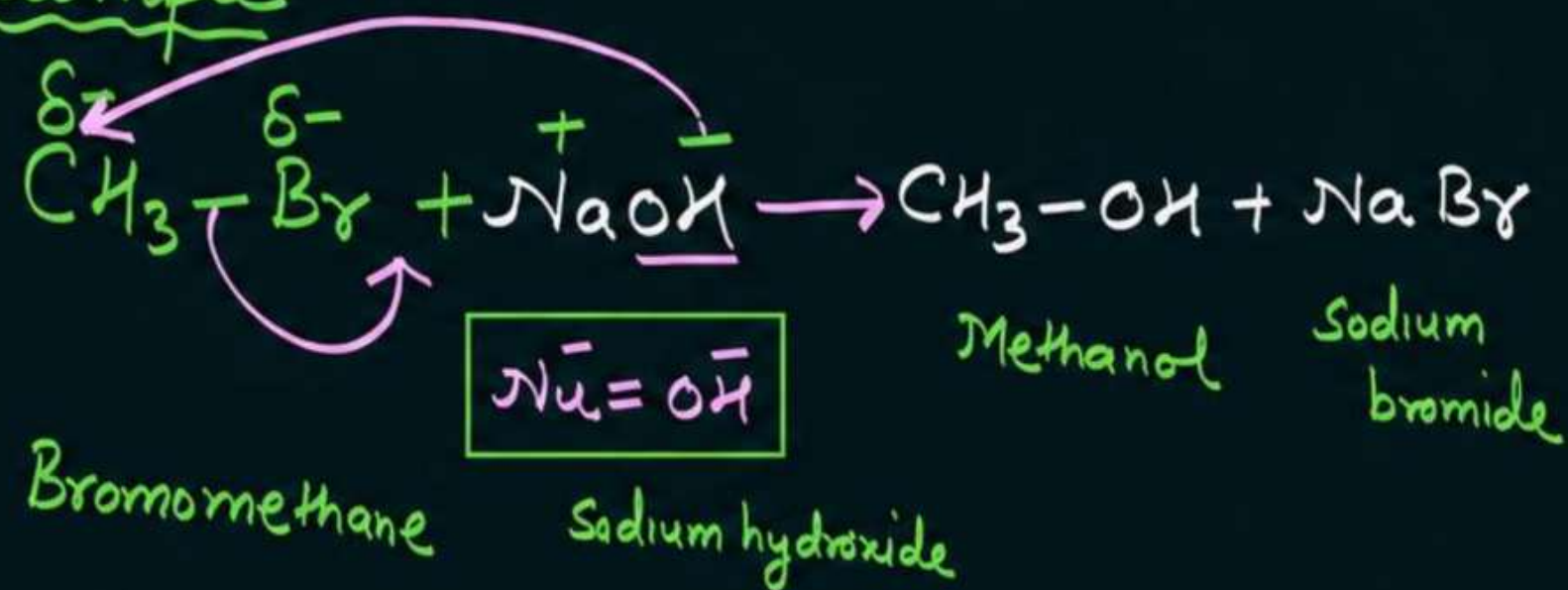
1. Substitution RxN

The reactions in which an atom or group of atoms in a molecule is replaced or substituted by different atoms or group of atoms are called substitution reactions.

① Nucleophilic Substitution Rxn.

Type of a substitution Rxn, which is initiated by nucleophile is called nucleophilic substitution Rxn.

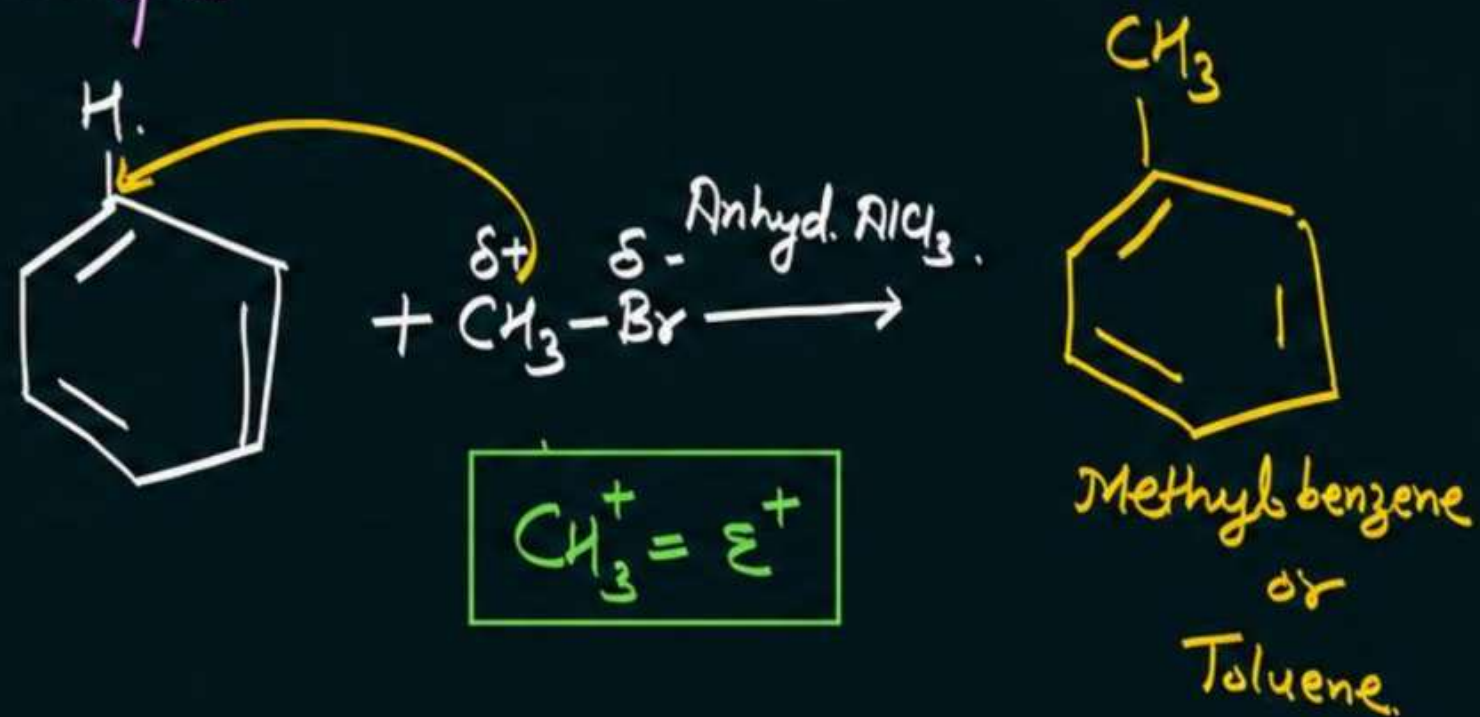
example



2) Electrophilic Substitution Rxn.

Type of a substitution Rxn, which is initiated by electrophile is called electrophilic substitution Rxn.

example



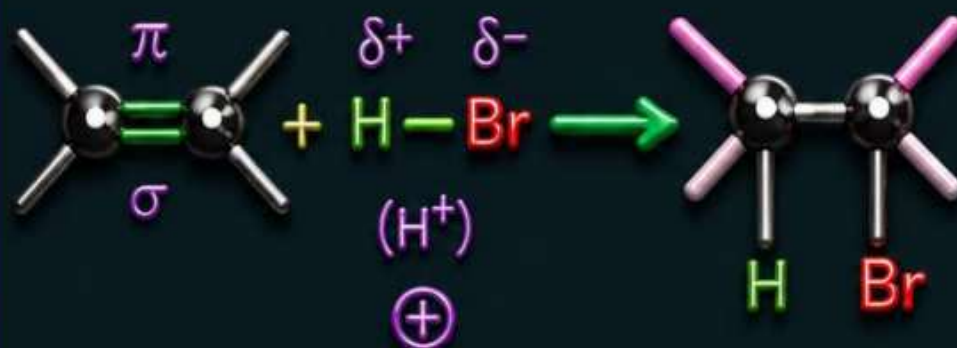
Types of Reactions

2. Addition reactions

The reactions in which an atom or group of atoms in a molecule is replaced or substituted by different atoms or group of atoms are called substitution reactions.

a) ELECTROPHILIC ADDITION RXN.

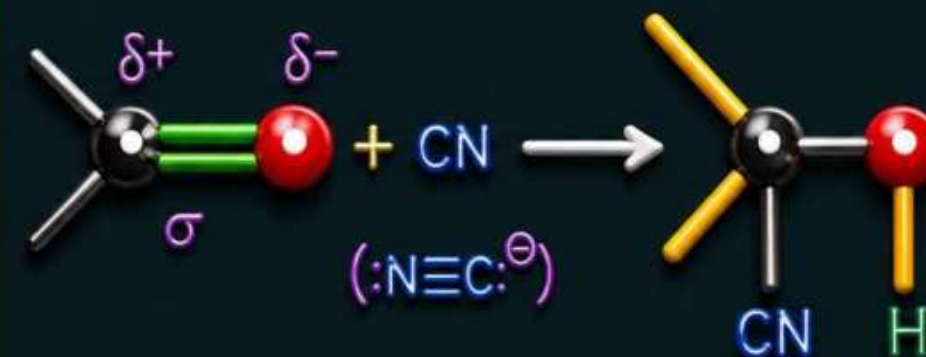
Type of addition Rxn which is initiated by electrophile. *e.g.*



- 1 π bond break
- 2 σ -bond formation

b) NUCLEOPHILIC ADDITION RXN.

Type of addition Rxn, which is initiated by nucleophile. *e.g.*



- 1 π -bond break
- 2 σ -bond formation

Types of Reactions

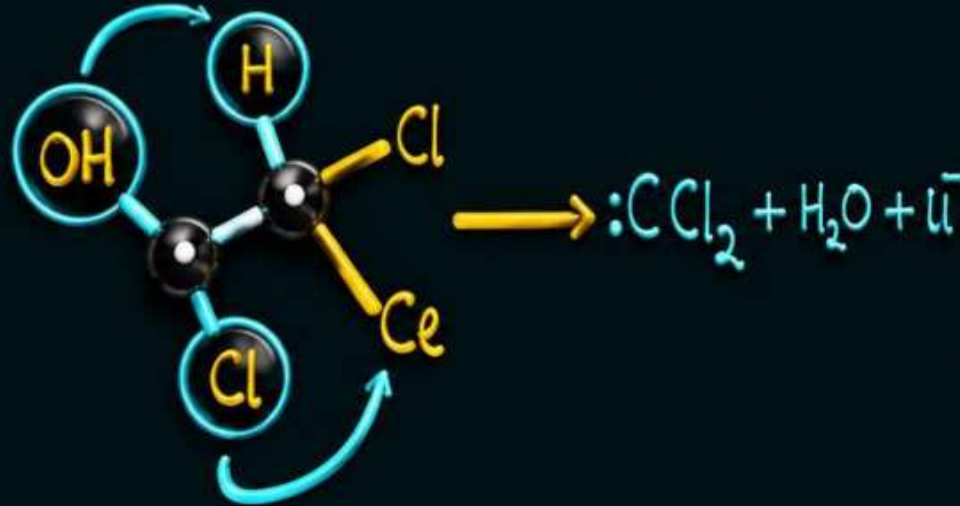
3. Elimination reactions

The reactions in which two atoms or groups of the molecule are removed are called elimination reactions.

For example,

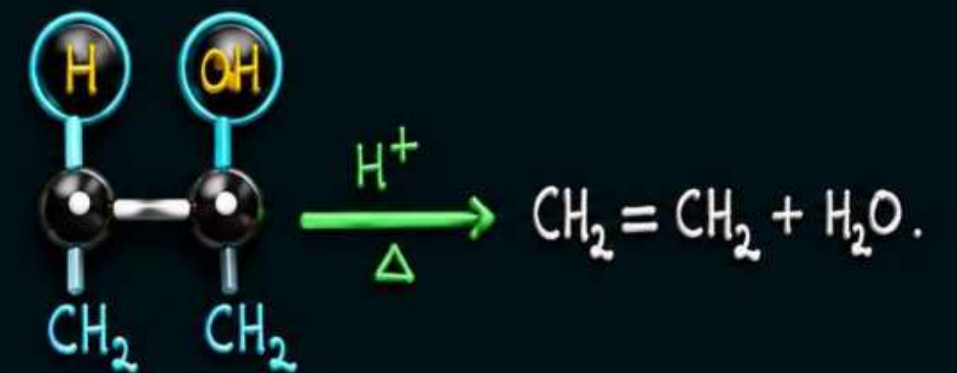
a) α -Elimination Rxn.

Type of elimination in which two groups or atoms are removed from same atoms e.g.



b) β -Elimination Rxn.

Type of elimination in which two atoms or groups are removed from adjacent C-atoms

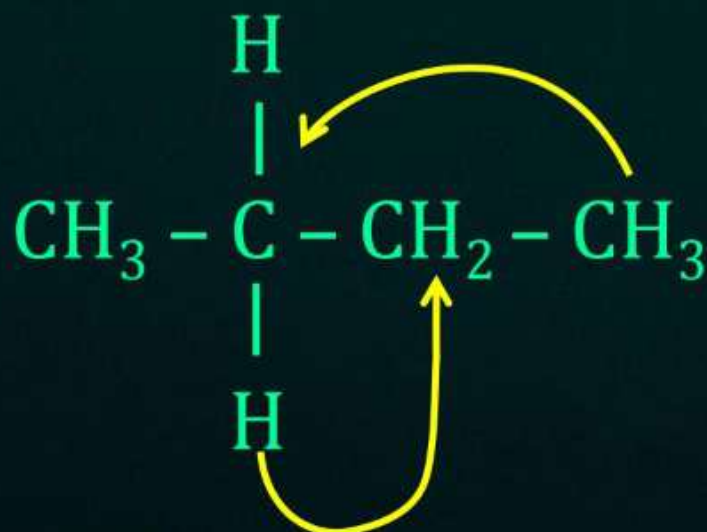
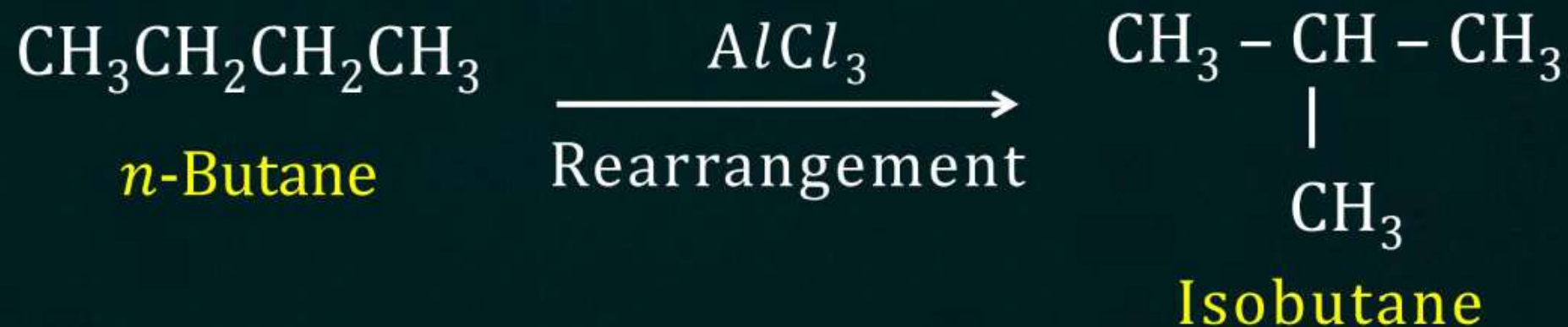


Types of Reactions

4. Rearrangement reactions

These reactions involve the migration of atoms or group of atoms to another position within the molecule under suitable conditions.

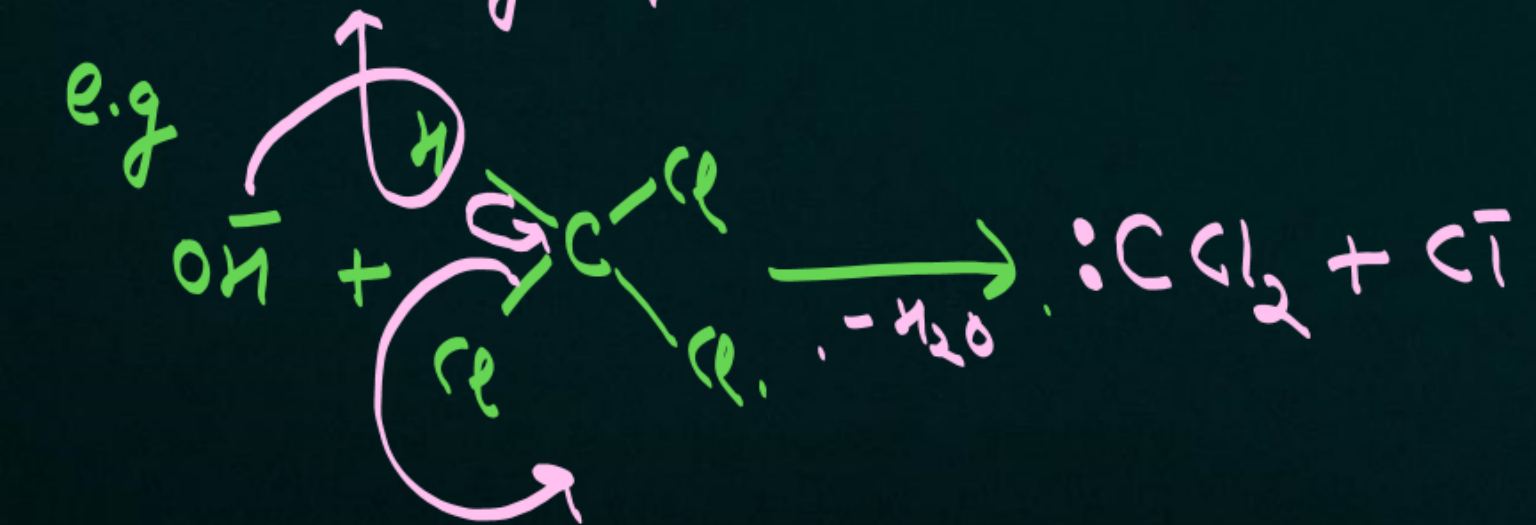
For example,



3). Elimination Reaction $\Rightarrow$ Type of organic reactions in which removal of one or two groups or atom takes place. is called Elimination Reaction.

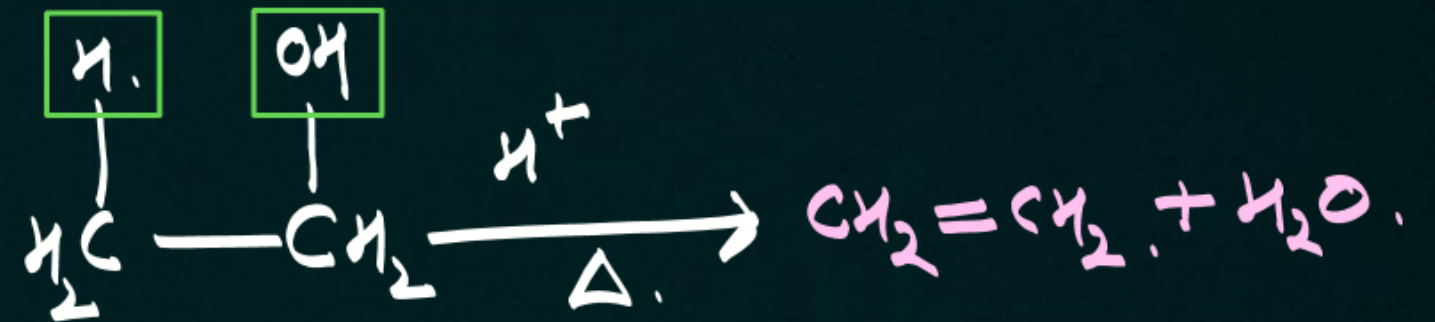
Two types $\Rightarrow$

a) α -Elimination $\Rightarrow$ If the removal of a atom or group takes place from same atom.

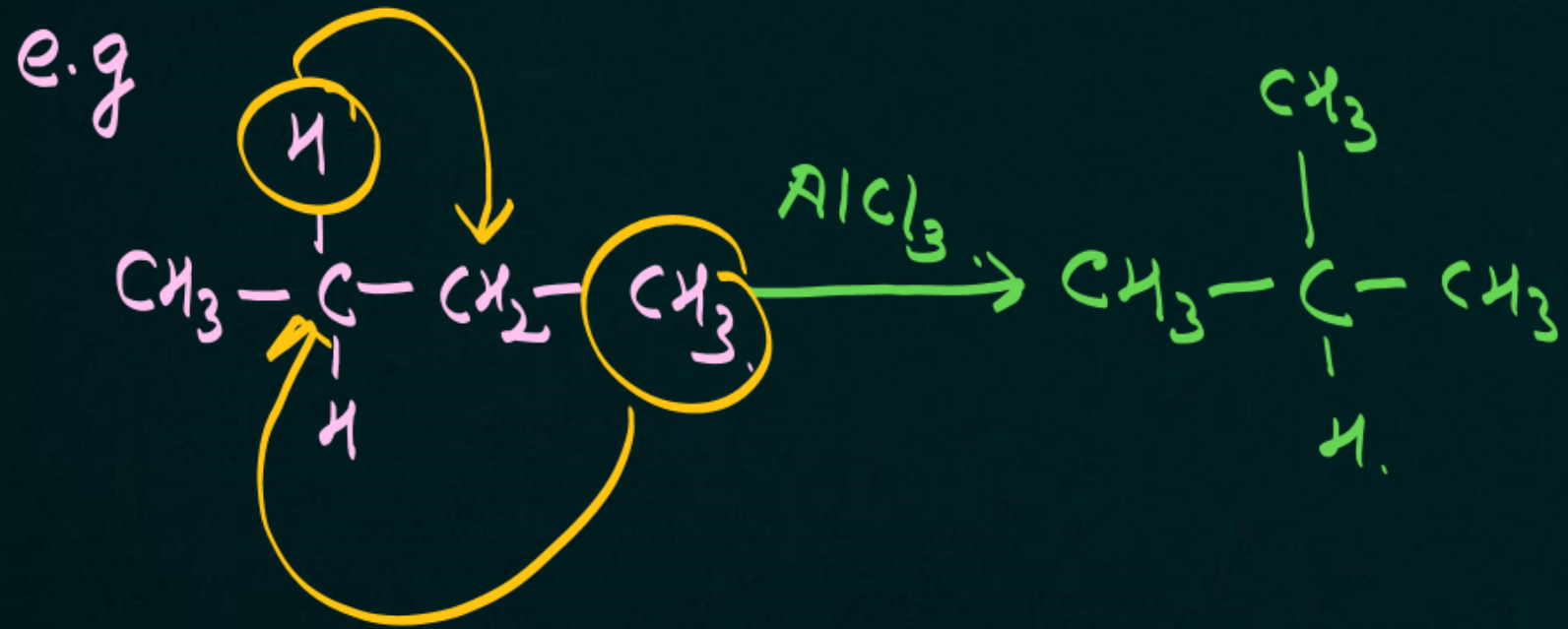


b) β -elimination rxn $\Rightarrow$

Removal of atoms or groups takes place from adjacent atoms.

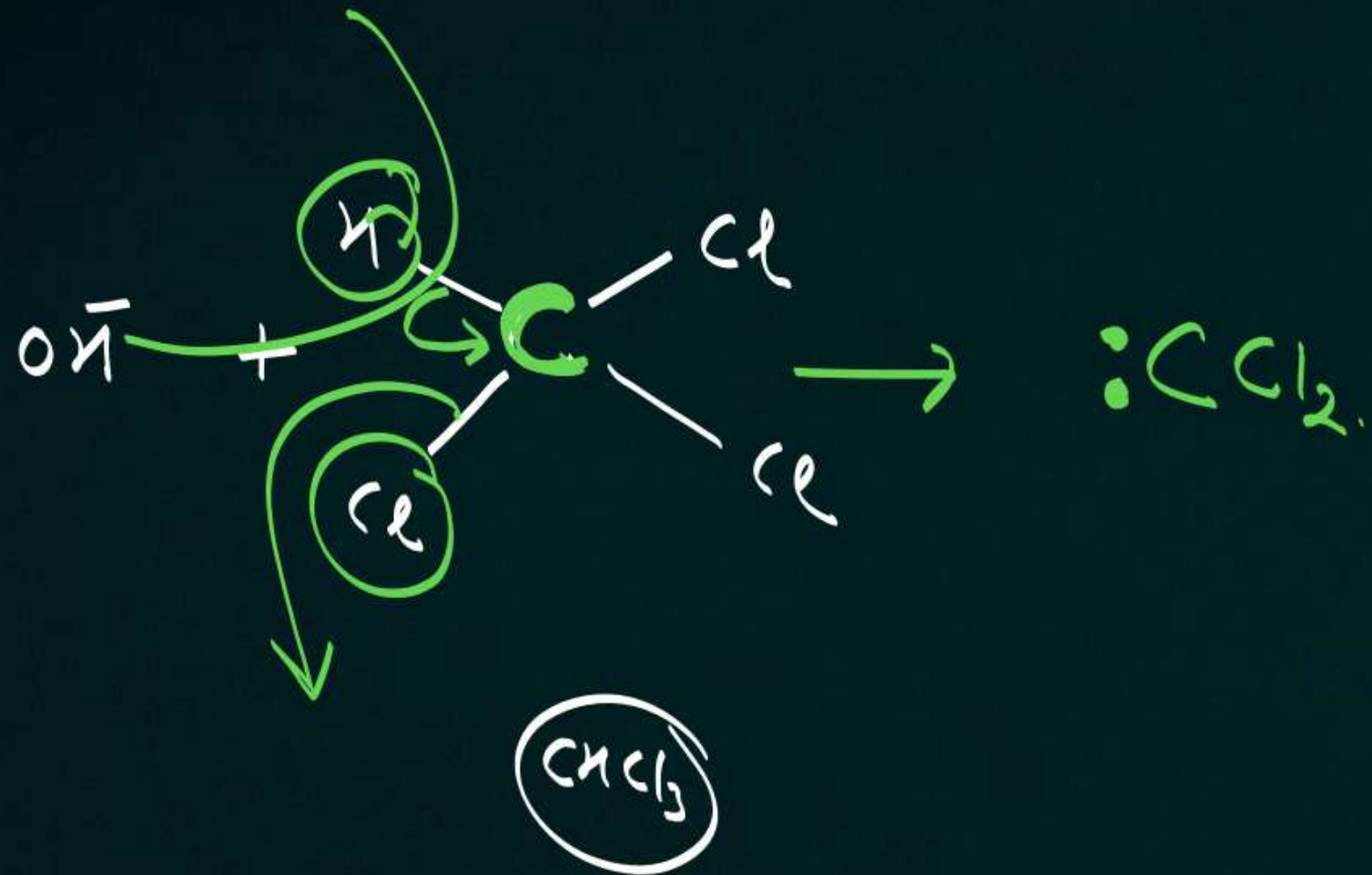


4) Rearrangement Reaction: $\rightarrow$ Type of organic reactions in which migration of atom or group of atoms takes place from one carbon to another carbon is called rearrangement reaction.

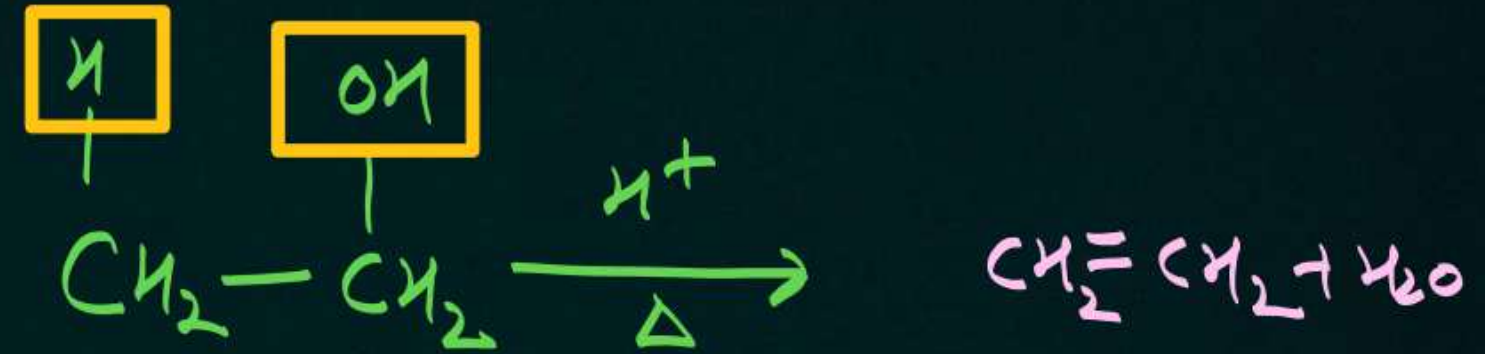


- 1) Addition
- 2) Substitution
- 3) Elimination
- 4) Rearrangement

$\alpha -$



β



Methods of Purification of Organic Compounds

- Once an organic compound is extracted from a natural source or synthesized in the laboratory, it is essential to purify it. Various methods used for the purification of organic compounds are based on the nature of the compound and the impurity present in it.

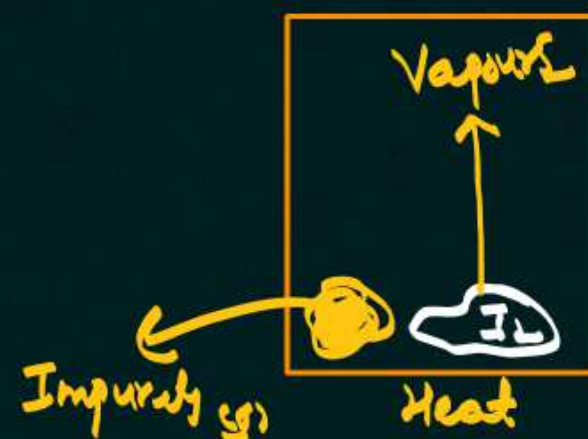
The common techniques used for purification are as follows:

- (i) Sublimation (Solid-vapour)
- (ii) Crystallisation M.P. (Solid → liq.)
- (ii) Distillation BP (liq → gas)
- (iv) Differential extraction and (Solubility)
- (v) Chromatography

Methods of Purification of Organic Compounds

1. SUBLIMATION: (Solid $\rightarrow$ Vapour)

- In sublimation, a solid is converted directly into gaseous state on heating and vice versa in case of cooling.



- For example, camphor, naphthalene, benzoic acid, iodine, ammonium chloride, anthracene, anthroquinone and phthalic anhydride are purified by sublimation.
- It is suitable for the solids which have non-volatile impurities and undergo sublimation on heating

↓
Impurity that does not vaporise.

Methods of Purification of Organic Compounds

2. CRYSTALLISATION: (M.P.) (Solid → Liq.)

Impure compound + solvent $\xrightarrow{(s)}$ Dissolve + crystals

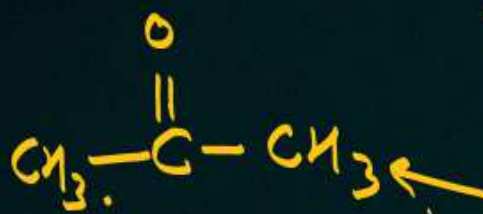
- Crystallization is based on the fact that during the crystal formation, impurities are left out as they are insoluble or feebly soluble in the used solvent.

For crystallization, a suitable solvent like H_2O , $R-OH$, $R-O-R$, CCl_4 , C_6H_6 , acetone or petroleum are used. For example,

- A mixture of benzoic acid and naphthalene is separated by hot water in which only benzoic acid dissolves.
- The impurities of common salt in sugar can be removed by hot ethanol as only sugar dissolves in it.

Fractional Crystallization (Repeatative Crystallisation)

- Fractional crystallization involves repeated crystallization.
- It is used when the substances differ slightly in the solubilities.
- For example, in a mixture of $KClO_3$ and KCl , KCl is more soluble than $KClO_3$ in cold water.



Methods of Purification of Organic Compounds

3. DISTILLATION: (B.P.) (Liq. $\rightarrow$ gas)

Impure compound (l) $\xrightarrow{\text{heat}}$ Vapour $\xrightarrow{\text{cold}}$ condensation

- The vapourisation of a liquid by heating and subsequent condensation of vapours by cooling is known as distillation. Distillation is an important method used to separate ~~and~~ its ~~principle~~ ~~work on~~:

- (a) Volatile liquids from non-volatile impurities.
- (b) Liquids having sufficient difference in boiling points.

- Distillation can be further divided into

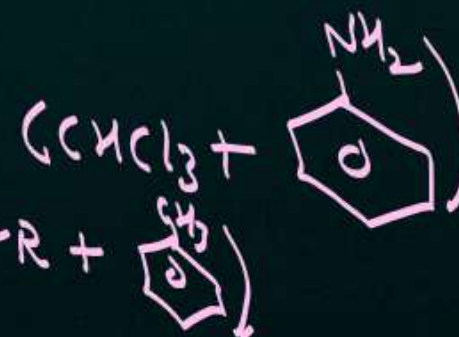
- i) do not decompose on bp.
- ii) non-volatile impurity
- iii) B.P difference 30-50K

(a) SIMPLE DISTILLATION (B.P. difference is sufficient)

- The substances which do not decompose on heating at their boiling point and have non-volatile impurities are purified by this method. It is used when liquids differ in their boiling point by 30-50 K.

➤ For example

- (i) Chloroform (B.P. 334 K) & Aniline (B.P. 457 K)
- (ii) Ether (B.P. 308 K) & Toluene (B.P. 384 K)
- (iii) Benzene (B.P. 353 K) & Aniline (B.P. 475 K)



Mixture is heated,
 volatile substance
 vapour easily, vapours
 are caught & are
 condensed in different
 containers. Non-
 volatile impurity
 remains in container
 which is being heated.

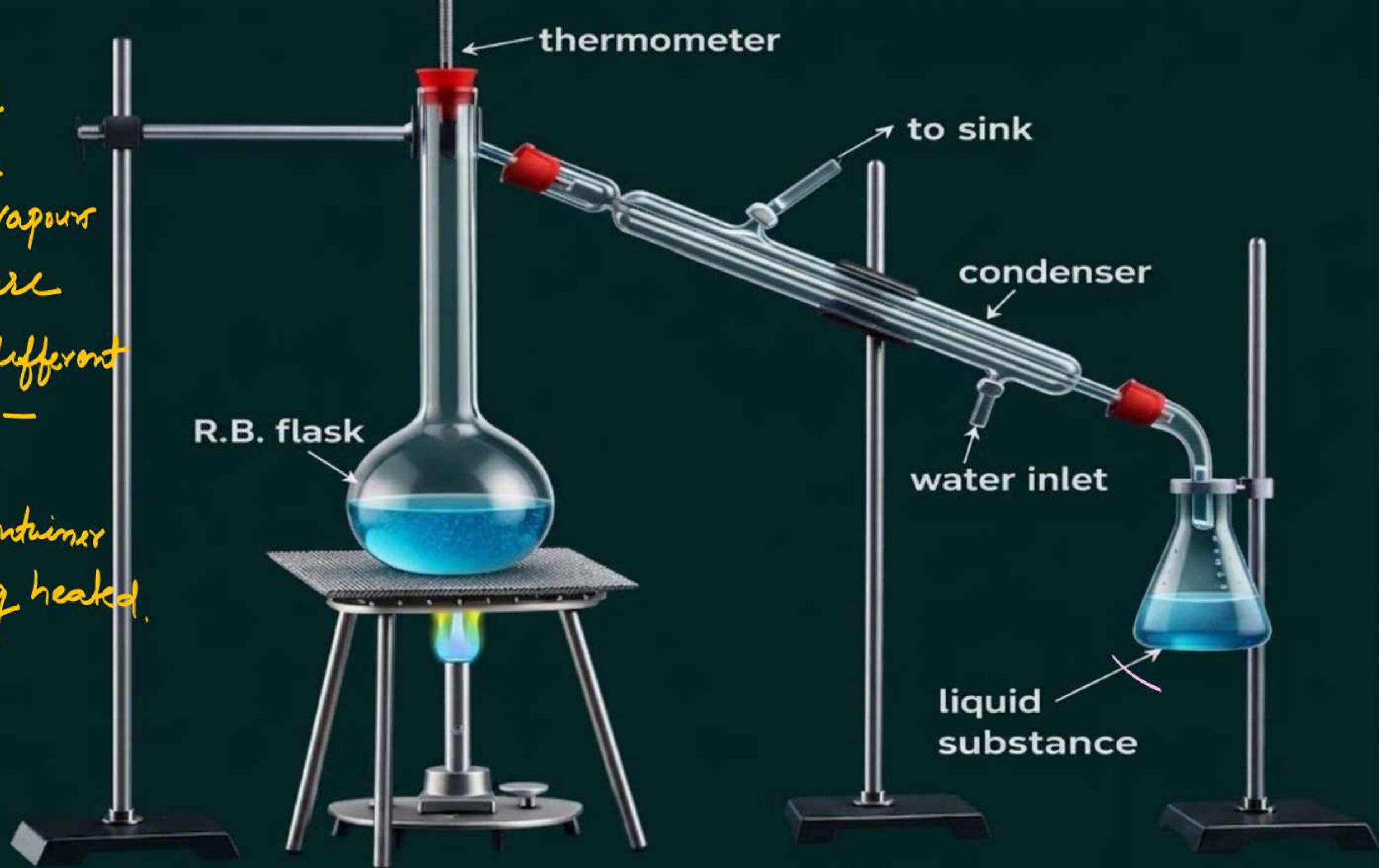


Figure: Simple distillation. The vapours of a substance formed are condensed and the liquid is collected in conical flask.

Methods of Purification of Organic Compounds

(b) FRACTIONAL DISTILLATION: (Very on difference in BP)

- Fractional distillation is used when the substances have nearly same boiling points **(5°C to 10°C difference only)**. It involves repeated distillations and condensations.
- This process is carried out by using fractionating columns. Fractionating column is a special type of long glass tube provided with obstructions to the passage of the vapour upwards and that of liquid downwards.
- The more volatile liquid (with a lower boiling point) distils off before the less volatile liquid (with higher boiling point).
- This method may be **used to separate a mixture of acetone (b. p. 330. K) and methyl alcohol (b. p. 338 K) or a mixture of benzene and toluene.** One of the technological applications of fractional distillation is to **separate different fractions of crude oil in petroleum industry into various useful fractions such as gasoline, kerosene oil, diesel oil and lubricating oil etc.**

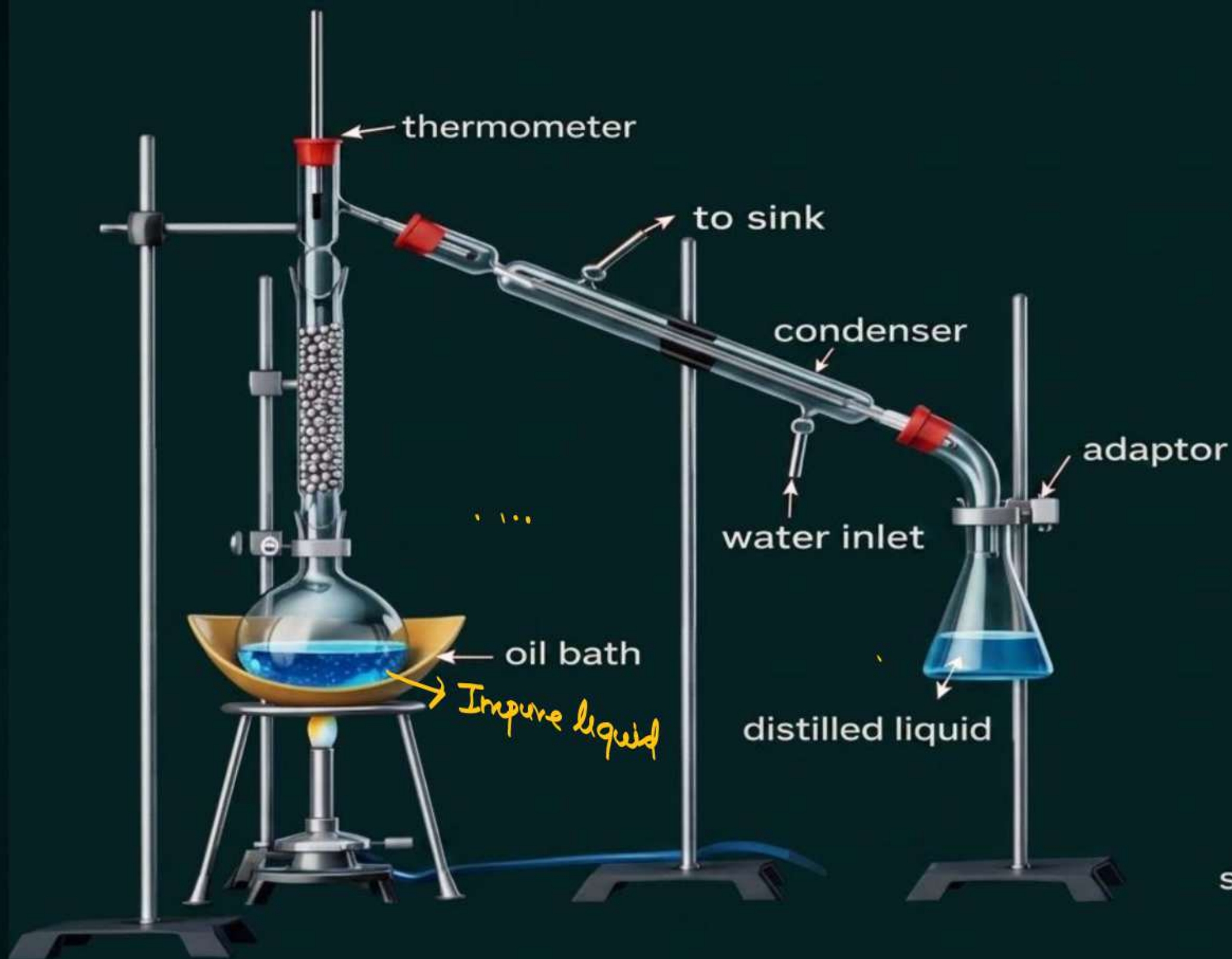


Figure: Fractional distillation. The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.

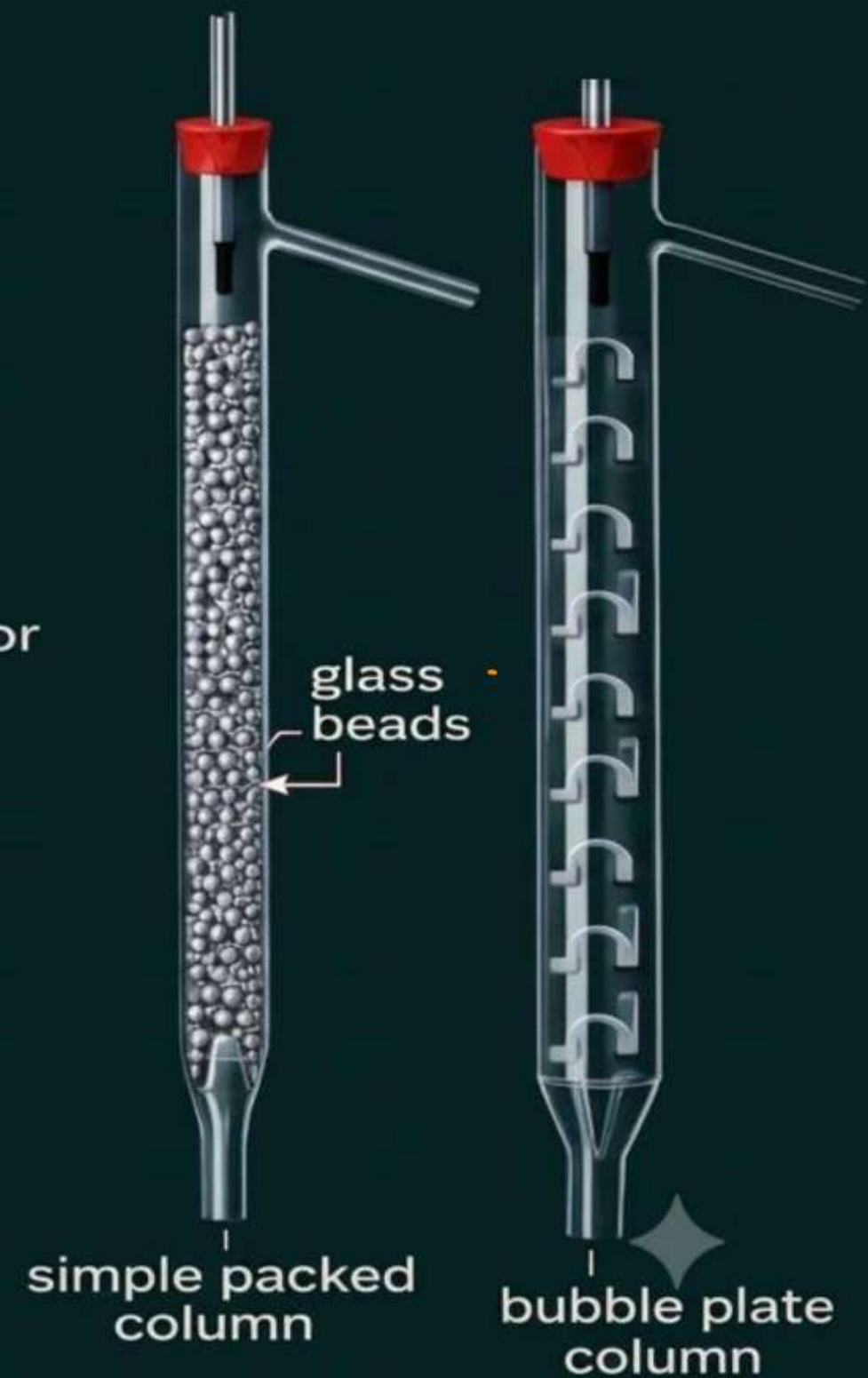


Figure: Different types of fractionating columns.

Methods of Purification of Organic Compounds

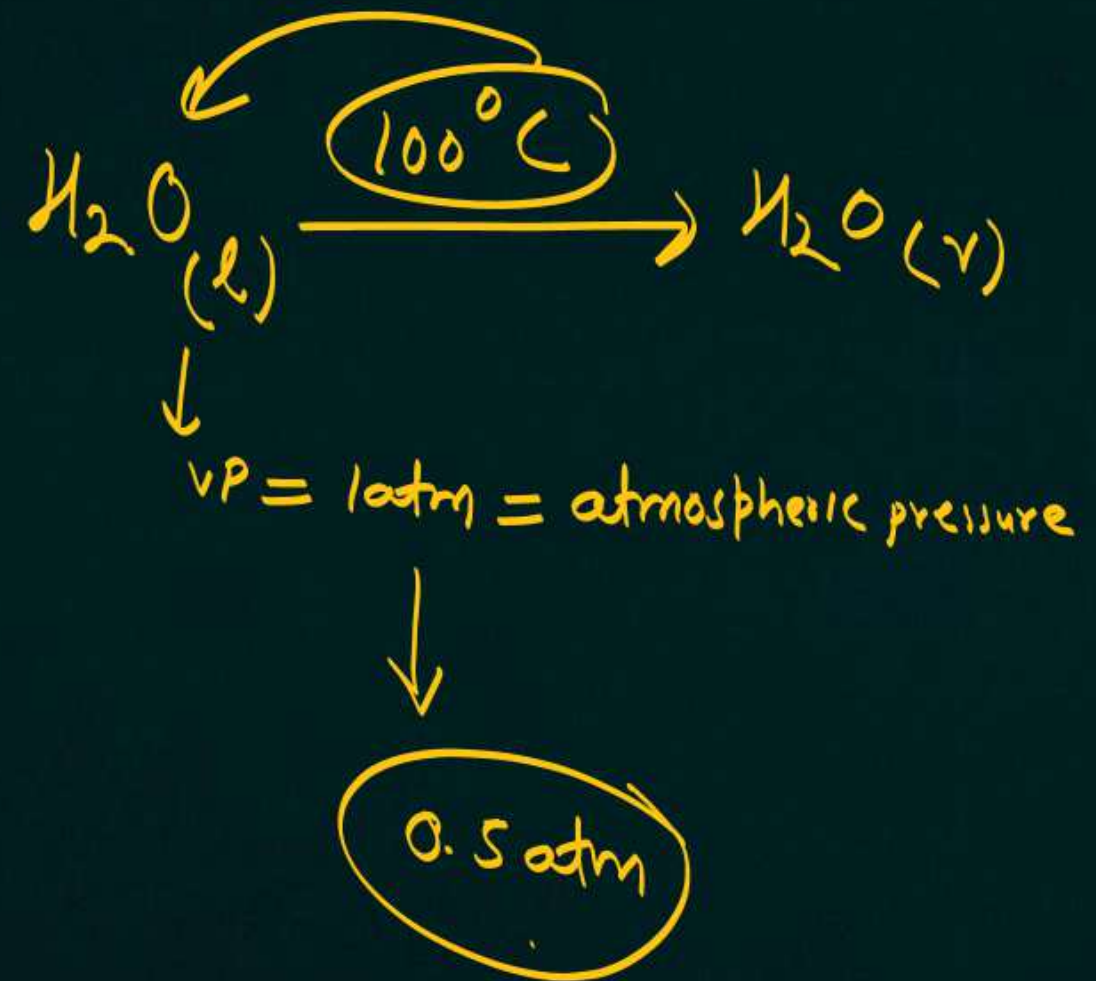
(c) DISTILLATION UNDER REDUCED PRESSURE: (BP is very high)

- This method is used to purify liquids having very high boiling points, which decompose at or below their boiling points.
- It is based on the assumption that a liquid boils at a temperature when its V.P. becomes equal to that of the atmospheric pressure. It means if pressure is reduced distillation can occur below the Boiling Point of the substance.
- Here the pressure is reduced during distillation to avoid decomposition, so it is also called distillation at reduced pressure.
- Glycerine is purified by vacuum distillation.
- H_2O_2 , formaldehyde are purified by vacuum distillation.
- Glycerol can be separated from **spent-lye** in soap industry by using vacuum distillation.
- Sugarcane juice is concentrated in sugar industry by evaporation under reduced pressure which saves lot of fuel.



@ Vish. de maar

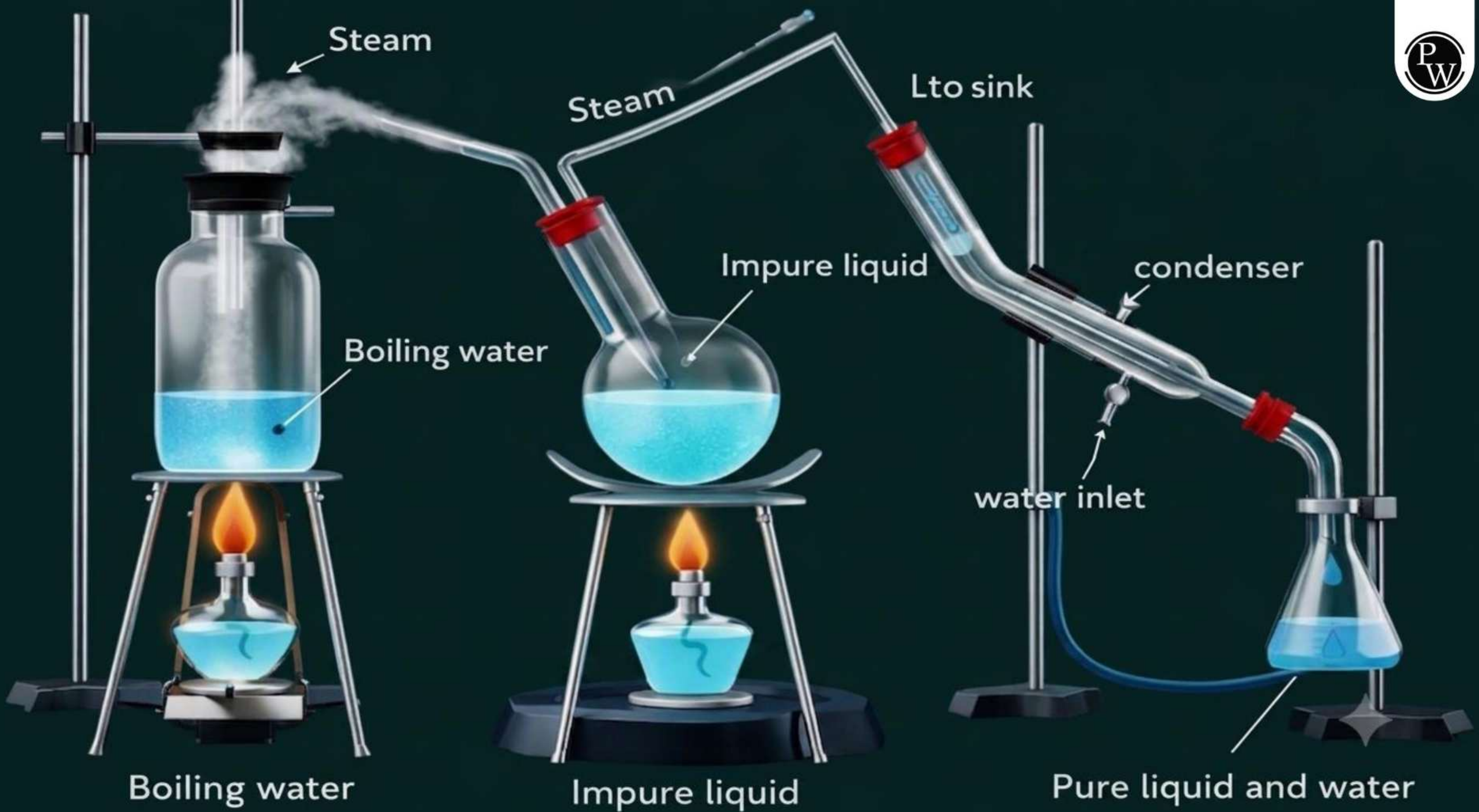
Three hand-drawn white arrows originate from the right side of the orange box, pointing towards the right edge of the slide.



Methods of Purification of Organic Compounds

(d) STEAM DISTILLATION: (BP is very low) (Those liquids which are water insoluble)

- This method is used for separation and purification of organic compounds (solids or liquids) which
 - (a) Are steam volatile
 - (b) Are insoluble in water.
 - (c) Possess high vapour pressure (10–15 mm of Hg at 373 K)
 - (d) Contains non volatile impurities.
- Compounds which can be purified by steam distillation are aniline, nitrobenzene, bromobenzene, *o*-nitrophenol, *o*-hydroxy benzaldehyde (salicylaldehyde), *o*-hydroxy acetophenone, turpentine oil and essential oils.



Methods of Purification of Organic Compounds

DIFFERENTIAL EXTRACTION: (Solubility)

- The process of isolating an organic compound from its aqueous solution by shaking with a suitable solvent is called differential extraction. It is also called solvent extraction.
- When an organic compound is present in an aqueous medium, then it is separated by shaking it with an organic solvent in which it is more soluble than in water.
- Solvent should be immiscible with water and organic compound to be separated should be highly soluble in it.
- Organic solvent and aqueous solution are immiscible with each other, so they can form two distinct layers which can be separated by separatory funnel.
- Organic solvent is distilled or evaporated to get organic compound.

Methods of Purification of Organic Compounds

- Organic compound is less soluble in organic solvent so large quantity of solvent is required to extract small quantity of compound, which is said to be continuous extraction. Benzoic acid can be extracted from its aqueous solution using benzene as solvent.
- Ether is a better solvent in differential extraction due to:
 - (a) Its less polarity
 - (b) Least reactivity
 - (c) Higher solubility of organic compounds

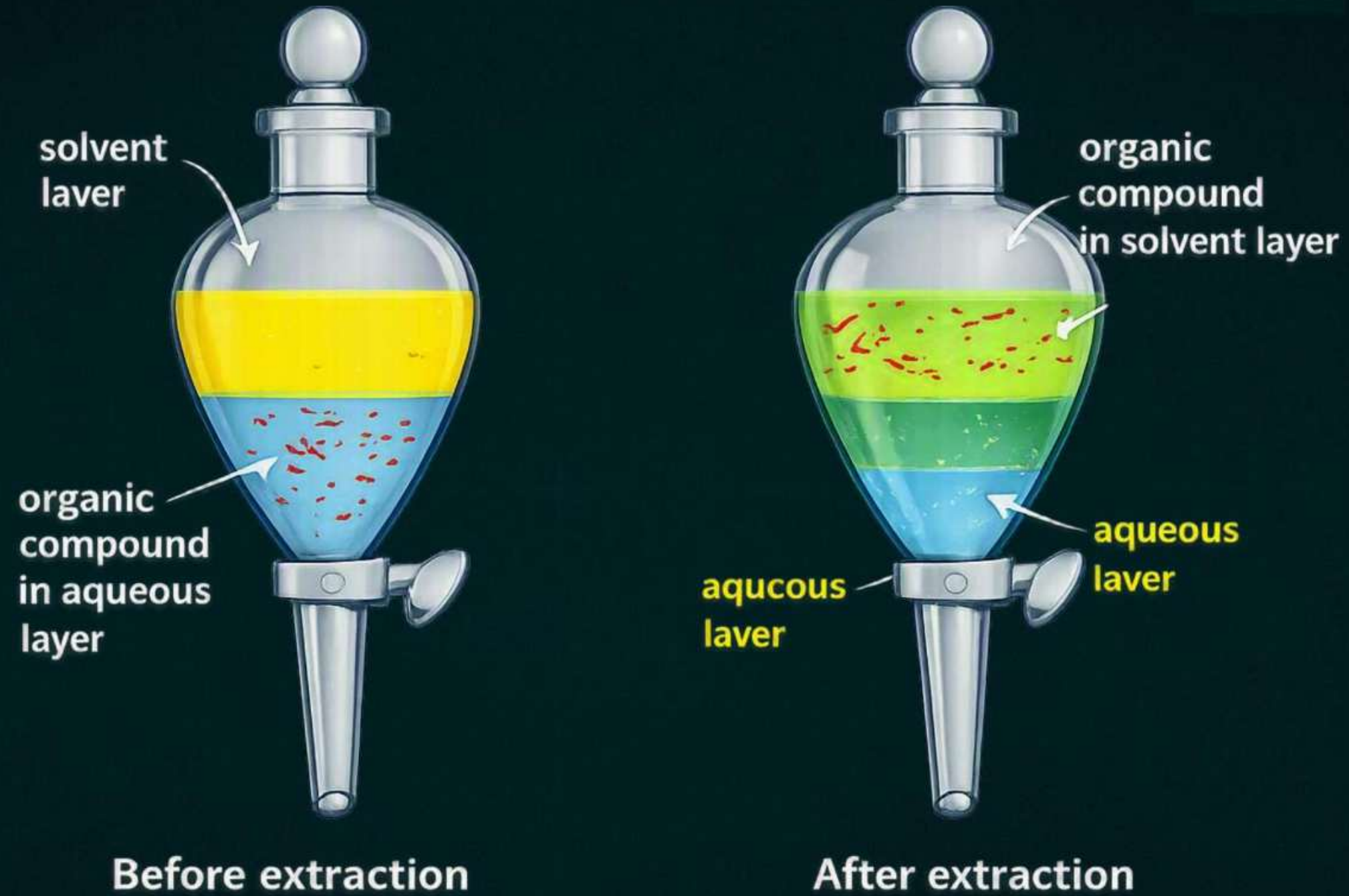


Figure: Differential extraction. Extraction of compound takes place based on difference in solubility

QUESTION



Sublimation is process where a solid:

- 1 Melts
- 2 Changes into liquid form.
- 3 Boils
- 4 Changes into vapour form directly.

QUESTION



Steam distillation is applied to those organic compounds which are steam volatile and:

- 1 Soluble in water.
- 2 Insoluble in water.
- 3 Sparingly soluble in water.
- 4 Insoluble in all solvents.

QUESTION



A fractional column is used in:

- 1 Sublimation
- 2 Distillation
- 3 Fractional distillation
- 4 Chromatography

QUESTION



Anthracene is purified by

- 1 Filtration
- 2 Distillation
- 3 Crystallisation
- 4 Sublimation.

BYEE JIGAR.

Thank
You

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♡